1.7 TYPES OF LATTICES

As described earlier, the number of point groups in two and three dimensions are 10 and 32 respectively. These point groups form the basis for construction of different types of lattices. Only those lattices are permissible which are consistent with the point group operations. Such lattices are called *Bravais lattices.* It is beyond the scope of this book to describe the details of formation of various Bravais lattices from the possible point group operations. It can be stated that 10 and 32 point groups in two and three dimensions produce only 5 and 14 distinct Bravais lattices respectively. These Bravais lattices further become parts of 4 and 7 distinct *crystal systems* respectively and are separately described below.

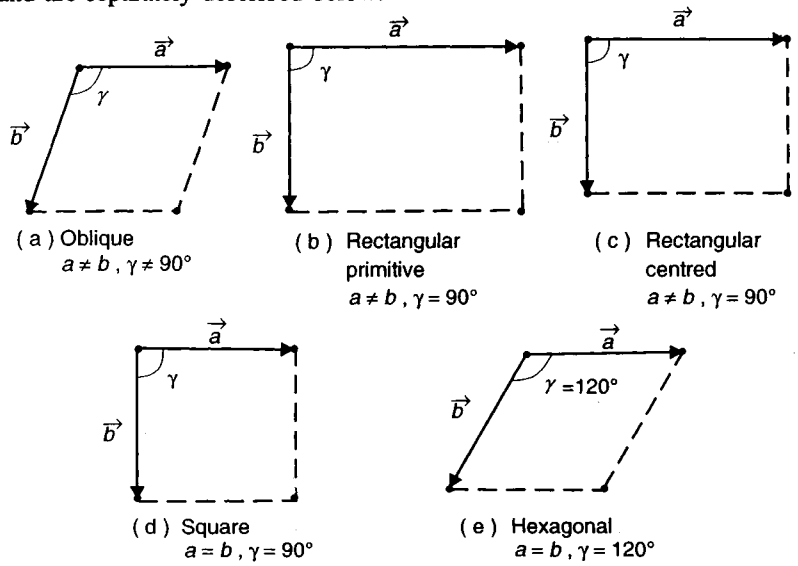


Fig. 1.10. *Bravais lattices in two dimensions.*

(i) Two-Dimensional Lattices

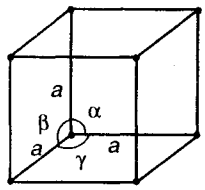
The four crystal systems of two-dimensional space are oblique, rectangular, square and hexagonal. The rectangular crystal system has two Bravais lattices, namely, rectangular primitive and rectangular centred. In all, there are five Bravais lattices which are listed in Table 1.1 along with the corresponding point groups. These lattices are shown in Fig. 1.10.

TABLE 1.1. Crystal systems and Bravais lattices in two dimensions

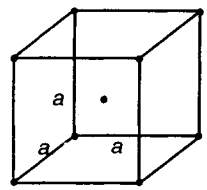
S. No.	Crystal system	Characteristic point group symmetry	Bravais lattice	Conventional unit cell	Unit cell characteristics
1	Oblique	1, 2	Oblique	Parallelogram	$a \neq b, \gamma \neq 90^0$
2	Rectangular	1*m*, 2*mm*	1. Rectangular primitive 2. Rectangular centred	Rectangle	$a \neq b, \gamma = 90^o$
3	Square	4, 4*mm*	Square	Square	$a = b, \gamma = 90^o$
4	Hexagonal	3, 3*m*, 6, 6*mm*	Hexagonal	60° Rhombus	$a = b, \gamma = 120^o$

(ii) Three-Dimensional Lattices

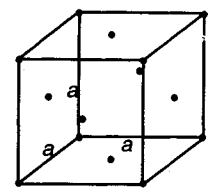
All the seven crystal systems of three-dimensional space and the corresponding Bravais lattices are listed in Table 1.2 in the decreasing order of symmetry. The crystallographic axes **a, b** and **c** drawn from one of the lattice points determine the size and shape of a unit cell. The angles α, β and γ represent the angles between the vectors **b** and **c, c** and **a,** and **a** and **b** respectively. The lengths *a*, *b* and *c* and angles α, β and γ are collectively known as *lattice parameters* or *lattice constants* of a unit cell. These Bravais lattices are also shown in Fig. 1.11 in the form of their conventional unit cells. The symbols P, F and I represent simple or primitive, face-centred, and body-centred cells respectively. A base or end-centred cell is that which has lattice points at corners and at one of the pairs of opposite faces. It is designated by the letter A, B or C. The designation A refers to the cell in which the faces defined by **b** and **c** axes contain the lattice points, and so on. The symbol R is specifically used for rhombohedral lattice.



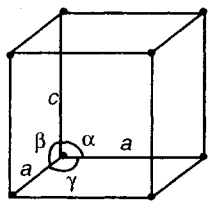
Simple cubic (P)



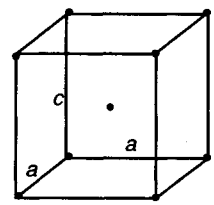
Body – centred cubic (I)



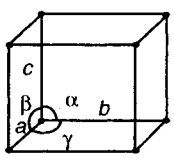
Face – centred cubic (F)



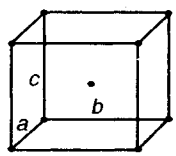
Simple tetragonal (P)



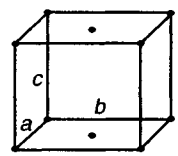
Body – centred tetragonal (I)



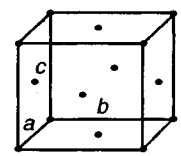
Simple orthorhombic (P)



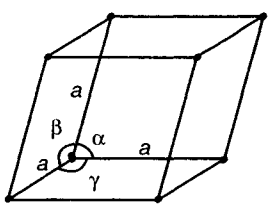
Body – centred orthorhombic (I)



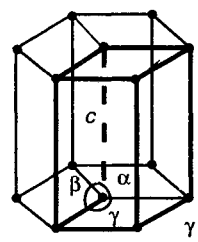
End – centred orthorhombic (C)



Face – centred orthorhombic (F)

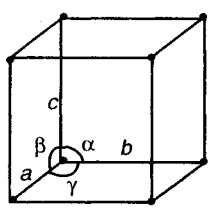


Simple rhombohedral (R)

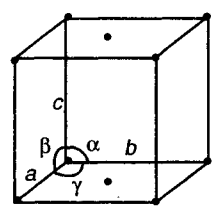


γ = 120 °

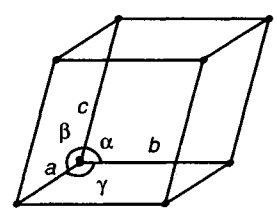
Simple hexagonal (P)



Simple monoclinic (P)



End – centred monoclinic (P)



Simple triclinic (P)

Fig.1.11. *The Bravais lattices in three dimensions*

TABLE 1.2. Crystal systems and Bravais lattices in three dimensions

S. No.	Crystal system	Lattice parameters	Bravais lattice	Common abbreviation	Lattice symbol	Examples
1	Cubic	$a = b = c$ $\alpha = \beta = \gamma = 90^\circ$	Simple Body-centred Face-centred	sc bcc fcc	P I F	Cu, Ag, Fe, Na, NaCl, CsCl
2	Tetragonal	$a = b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	Simple Body-centred	st bct	P I	β-Sn, TiO_2
3	Orthorhombic	$a \neq b \neq c$ $\alpha = \beta = \gamma = 90^\circ$	Simple Body-centred End-centred Face-centred	so bco eco fco	P I C F	Ga, Fe_3C (cementite)
4	Rhombohedral or Trigonal	$a = b = c$ $\alpha = \beta = \gamma \neq 90^\circ$	Simple	–	P (or R)	As, Sb, Bi
5	Hexagonal	$a = b \neq c$ $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$	Simple	–	P	Mg, Zn, Cd, NiAs
6	Monoclinic	$a \neq b \neq c$ $\alpha = \gamma = 90^\circ \neq \beta$	Simple End-centred	– –	P C	$CaSO_4 . 2H_2O$ (gypsum)
7	Triclinic	$a \neq b \neq c$ $\alpha \neq \beta \neq \gamma \neq 90^\circ$	Simple	–	P	$K_2Cr_2O_7$

The hexagonal crystal system has only one Bravais lattice and its unit cell may be either of cubical or of hexagonal type. The cubical type cell (outlined by thick lines) has lattice points only at the corners.The hexagonal cell has lattice points at the corners as well as at the centres of the two hexagonal faces. One hexagonal cell is formed by joining together three cubical type cells.

A lattice point lying at the corner of a cell is shared by eight such cells and the one lying at the face centre position is shared by two cells. Therefore, the contribution of a lattice point lying at the corner towards a particular cell is 1/8 and that of a point lying at the face centre is ½. The following equation is used to calculate the effective number of lattice points, N, belonging to a particular cell :

$$N = N_i + N_f / 2 + N_c/8 \qquad (1.5)$$

where N_i represents the number of lattice points present completely inside the cell, and N_f and N_c represent the lattice points occupying face centre and corner positions of the cell respectively. Using this relation, the effective number of lattice points in a simple cubic, body-centred cubic and face-centred cubic lattices comes out to be 1, 2 and 4 respectively.

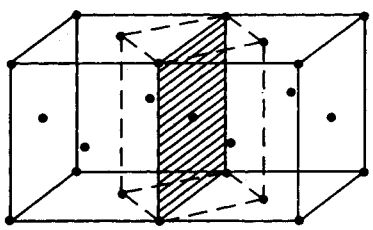


Fig. 1.12. *Two face-centred tetragonal lattices placed side by side result in an end-centred tetragonal lattice shown by dotted lines.*

The list of Bravais lattices given in Table 1.2 appears to be incomplete. The orthorhombic system contains four Bravais lattices whereas the cubic and tetragonal systems contain only three and two lattices respectively. It can be shown that the lattices which are absent in certain crystal systems do not result in new types of arrangements and so need not be considered separately. Figure 1.12 shows two face-centred tetragonal lattices placed side by side. This arrangement of points, shown by dotted lines, produces body-centred tetragonal lattice which already exists in the Bravais list.

1.8 LATTICE DIRECTIONS AND PLANES

The direction of a line in a lattice is defined by assigning certain indices to this line. If the line passes through the origin, its indices are determined by taking a point on this line and finding out the projections of the vector drawn from the origin to that point on the crystallographic axes. Let these projections be u, v and w. Apparently, u, v and w also represent the coordinates of that point. These coordinates are then simplified to get a set of the smallest possible integers which when enclosed in square brackets represent the indices of the line. As an example, to determine the indices of the direction OQ in a cubic crystal (Fig. 1.13), we may take either a point P (½, ½, ½) or Q (1, 1, 1) on this line; either of these points yields the indices of the direction OQ as [111]. The same are the indices of any other direction parallel to OQ because by shifting the origin to an appropriate position, the new direction can be made to pass through the points O, P and Q. The origin is shifted in such a way that the orientation of the axes remains unchanged. If a direction is perpendicular to a certain axis, its index corresponding to that axis is zero as it does not form any projection on the axis. A direction having projections on the negative sides of the axes possess negative indices which are written by putting bars over the indices.

Consider the direction AB as shown in Fig. 1.14. The indices of this

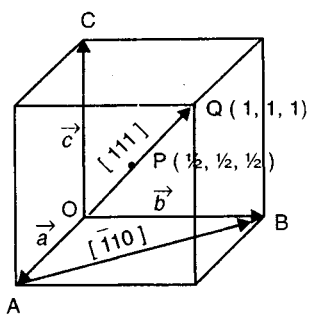


Fig. 1.13. *Determination of indices of a direction.*

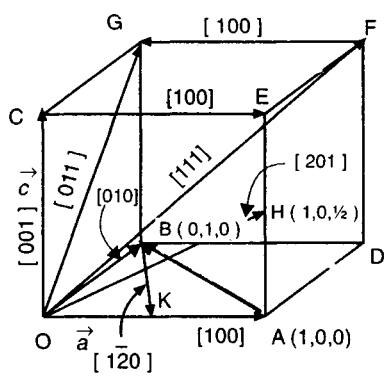


Fig. 1.14. *Indices of some directions in a cubic lattice.*

direction can be obtained by shifting the origin to the point A. The projections of AB on the axes then become −1, 1, and 0 and hence the indices of the line AB are [1$\bar{1}$0]. The indices of a few more directions are illustrated in Fig. 1.14. The cube edges are represented by the indices of the type [100], [010], [$\bar{1}$00], etc. These constitute a family of cube edges designated as <100> which includes all the directions of this type. Similarly, a

family of face diagonals is represented by <110> and that of body diagonals by <111>. The number of members in the families of cube edges, face diagonals and body diagonals is 6, 12 and 8 respectively.

The angle θ between the two crystallographic directions [*hkl*] and [*h'k'l'*] is given by

$$\cos\theta = \frac{hh' + kk' + ll'}{(h^2 + k^2 + l^2)^{1/2}\,(h'^2 + k'^2 + l'^2)^{1/2}} \qquad (1.6)$$

The scheme to represent the orientation of planes in a lattice was first introduced by Miller, a British crystallographer. The indices of planes are, therefore, known as the *Miller indices*. The steps involved to determine the Miller indices of a plane are as follows :

(i) Find the intercepts of the plane on the crystallographic axes.

(ii) Take reciprocals of these intercepts.

(iii) Simplify to remove fractions, if any, and enclose the numbers obtained into parentheses.

In step (i), the intercepts are taken in terms of the lengths of fundamental vectors choosing one of the lattice points as the origin. If a plane is parallel to a certain axis, its intercept with that axis is taken as infinity. In step (ii) the reciprocals are taken in order to avoid the occurrence of infinity in the Miller indices.

As an example, consider a plane ABC (Fig. 1.15) having intercepts 1, 2 and 1 with the crystallographic axes **a**, **b** and **c** respectively of a cubic lattice. The Miller indices of this plane are determined as follows :

(i) Intercepts : 1, 2, 1

(ii) Reciprocals : 1, ½, 1

(iii) Simplification : 2, 1, 2

Hence the Miller indices of the plane ABC are (212) ; the numbers within the parentheses are written without commas. The Miller indices of a plane, in general, are written as (*hkl*).

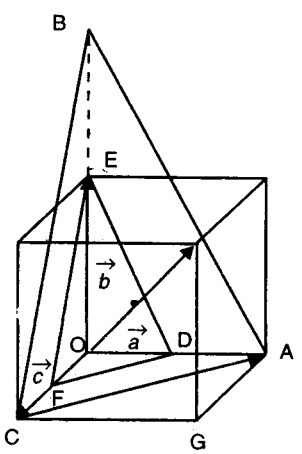


Fig. 1.15. *Miller indices of parallel planes and the planes passing through the origin.*

It may be noted that another plane DEF which is parallel to the plane ABC and lies completely inside the lattice, has intercepts 1/2, 1 and 1/2 with the

axes and hence carries the same Miller indices as the plane ABC. Thus we conclude that the parallel planes have the same Miller indices. The plane DEF is rather more convenient to deal with as it lies completely inside the lattice.

If a plane intercepts an axis on the negative side, a bar is put above the corresponding number of the Miller indices. The intercepts of a plane passing through the origin cannot be determined as such. In such a case, we take another plane parallel to this plane and determine its Miller indices. The same are the indices of the given plane. Alternatively, we shift the origin from the plane to some other suitable lattice point without changing the orientation of the axes and then find the Miller indices. For example, the indices of the plane OCGA in Fig. 1.15 become $(0\bar{1}0)$ if the origin is shifted to the point E. The importance of orientation of the axes can be realized with reference to Fig. 1.12. The indices of the shaded plane are of the type (100) when referred to the axes of the face-centred tetragonal cell, whereas these become of the type (110) when referred to the axes of the simple tetragonal cell indicated by dotted lines.

A family of planes of a particular type is represented by enclosing the Miller indices of any one of the planes of that family into braces. Thus {100} represents a family of planes which has the planes (100), (010), (001), $(\bar{1}00)$, $(0\bar{1}0)$ and $(00\bar{1})$ as its members. These six planes represent the faces of the cube. Similarly, the families of diagonal planes and close-packed planes are represented by {110} and {111}, and contain 6 and 8 members respectively. Some of these planes are illustrated in Fig. 1.16.

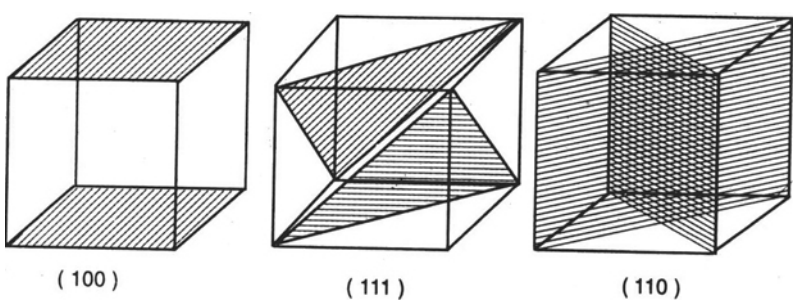


Fig. 1.16. *Two parallel planes belonging to each one of the families* {100}, {111} *and* {110} *in a cubic lattice.*

1.9 INTERPLANAR SPACING

Consider a set of parallel planes with indices (*hkl*). Take origin on a

lattice point of one such plane and draw crystallographic axes **a**, **b** and **c**. Now consider another similar plane adjacent to this plane. Since the second plane also has the Miller indices (*hkl*), the lengths of the intercepts on **a**, **b** and **c** are *a*/*h*, *b*/*k* and *c*/*l* respectively. If we draw a normal from the origin to the second plane, the length of the normal represents the interplanar distance *d*. From Fig. 1.17, it follows that

$$d = \text{OP} = \frac{a}{h}\cos\alpha = \frac{b}{k}\cos\beta = \frac{c}{l}\cos\gamma \tag{1.7}$$

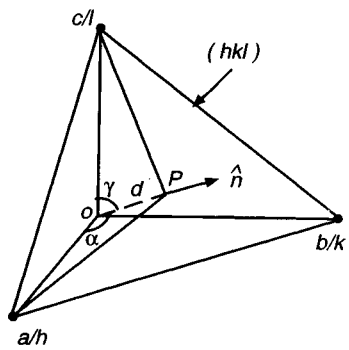


Fig. 1.17. *An (hkl) plane at a distance d from another similar parallel plane passing through the origin.*

where α, β and γ represent the angles between the normal and the axes **a**, **b** and **c** respectively and cos α, cos β and cosγ represent the direction cosines of the normal to the plane (*hkl*). The Eq. (1.7) indicates that the direction cosines of the normal are proportional to *h*/*a*, *k*/*b* and *l*/*c*.

If $\hat{\mathbf{n}}$ be the unit vector of the normal to the plane, then $a\cos\alpha$, $b\cos\beta$ and $c\cos\gamma$ may be written as $\hat{\mathbf{n}}.\mathbf{a}$, $\hat{\mathbf{n}}.\mathbf{b}$ and $\hat{\mathbf{n}}.\mathbf{c}$ respectively and Eqs. (1.7) become

$$d = \hat{\mathbf{n}}.\mathbf{a}/h = \hat{\mathbf{n}}.\mathbf{b}/k = \hat{\mathbf{n}}.\mathbf{c}/l \tag{1.8}$$

Thus the value of *d* can be determined if $\hat{\mathbf{n}}$ is known. In an orthogonal lattice, where **a**, **b** and **c** point along *x*, *y* and *z* directions respectively, the equation of the plane (*hkl*) with intercepts *a*/*h*, *b*/*k* and *c*/*l* on the axes is

$$f(x, y, z) = hx/a + ky/b + lz/c = 1$$

For a surface $f(x, y, z)$ = constant, ∇f represents the vector normal to it.

$$\therefore \quad \hat{\mathbf{n}} = \frac{\nabla f}{|\nabla f|} = \frac{(h/a)\hat{\mathbf{i}} + (k/b)\hat{\mathbf{j}} + (l/c)\hat{\mathbf{k}}}{(h^2/a^2 + k^2/b^2 + l^2/c^2)^{1/2}}$$

Hence from Eqs. (1.8), we obtain

$$\mathbf{d} = \frac{\hat{\mathbf{n}}.\mathbf{a}}{h} = \frac{\left[(h/a)\hat{\mathbf{i}} + (k/b)\hat{\mathbf{j}} + (l/c)\hat{\mathbf{k}}\right].(a/h)\hat{\mathbf{i}}}{(h^2/a^2 + k^2/b^2 + l^2/c^2)^{1/2}}$$

or

$$d = \frac{1}{(h^2/a^2 + k^2/b^2 + l^2/c^2)^{1/2}} \tag{1.9}$$